

Ultrasound-guided stellate ganglion block in the sitting position for insomnia treatment: a randomized controlled trial

Minqiang Liu

the Third People's Hospital of Shenzhen, The Second Affiliated Hospital of Southern University of Science and Technology

Liping Qiu

the Third People's Hospital of Shenzhen, The Second Affiliated Hospital of Southern University of Science and Technology

Aihua Chen

the Third People's Hospital of Shenzhen, The Second Affiliated Hospital of Southern University of Science and Technology

Xiaoliang Cha

the Third People's Hospital of Shenzhen, The Second Affiliated Hospital of Southern University of Science and Technology

Qiang Wu

45367039@qq.com

the Third People's Hospital of Shenzhen, The Second Affiliated Hospital of Southern University of Science and Technology

Article

Keywords: Stellate ganglion block, Insomnia, Sitting position, Patient comfort, Psychological stress

Posted Date: February 11th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-7783378/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

Background

Stellate ganglion block (SGB) is usually performed in a supine or lateral position; this is unsuitable for patients with mobility impairments. This study aimed to assess the clinical efficacy of ultrasound-guided SGB in the sitting position compared to supine position for insomnia management.

Methods

181 patients with refractory insomnia were randomized to receive 7 sessions of SGB in either the supine (group A, 86 cases) or sitting (group B, 95 cases) position. All patients received in-plane needle insertion ultrasound-guided technique. The primary outcomes included procedural difficulty and success rate; secondary outcomes encompassed patient comfort, psychological stress, sleep quality metrics, therapy efficacy, and adverse events.

Results

All SGB procedures were successfully completed with minimal technical difficulty, the operation time and the success rate was similar between groups ($P > 0.05$). Sitting position significantly offered better comfort and less psychological stress ($P < 0.05$). Both groups achieved better sleep quality and remarkable outcome ($P < 0.05$), but no significant differences were observed in sleep parameters or adverse events between groups ($P > 0.05$).

Conclusion

Ultrasound-guided sitting SGB provides superior patient comfort and psychological benefits while maintaining clinic efficacy and safety comparable to supine positioning in insomnia treatment.

Introduction

Stellate ganglion block (SGB) has emerged as a valuable therapeutic modality in pain medicine through autonomic nervous system modulation and local microcirculatory improvement^{1,2}. In recent years, more and more studies demonstrate the efficacy of SGB in insomnia treatment^{3~5}. The physiologic basis lies in suppression of sympathetic hyperactivation, restoration of autonomic balance, reduction of myofascial tension, and modulation of the sleep-wake cycle, thereby improving sleep quality^{6,7}. While the conventional supine or lateral positions remains standard practice^{8,9}, these may present challenges for patients with cardiopulmonary dysfunction, trauma-related limitations, especially for those with severe anxiety disorders or mobility impairments.

Recent advancements in ultrasound-guided techniques have enabled positional flexibility¹⁰; the sitting position is a promising alternative due to its practical advantages^{11,12}. However, rigorous comparative data regarding the equivalence of sitting versus supine SGB for insomnia management remain limited.

This randomized controlled trial systematically evaluates the comparative efficacy and safety of these two positioning approaches in patients with refractory insomnia.

Results

Baseline Characteristics

A total of 362 patients were initially screened in this study. Among these, 132 patients were excluded due to failure to meet inclusion criteria or fulfilment of exclusion criteria, and an additional 34 cases were excluded as they declined repeated SGB treatment. Following randomization, 15 participants were further excluded from the analysis due to discontinuation of interventions. Finally, 181 patients were statistically analysed, including 86 cases in group A and 95 cases in group B (Fig. 1). Each subject underwent 7 sessions of SGB, resulting in a cumulative total of 602 procedures for Group A and 665 for Group B. Baseline demographic characteristics were comparable between the two groups (Table 1).

Table 1
Baseline demographic characteristics

Item	Group A (<i>n</i> = 86)	Group B (<i>n</i> = 95)	χ^2/F	<i>P</i>
Sex (male/female)	30 / 56	32 / 63	0.029	0.865
Age (year)	48.8 ± 12.3	47.3 ± 13.9	3.915	0.450
Insomnia duration (month)	78(36, 120)	60(24, 120)	1.197	0.643
Operation time (min)	4.8 ± 0.9	4.8 ± 0.8	0.791	0.862
SGB success	594(98.7%)	659(99.1%)	0.526	0.468

Technical outcomes

All procedures were completed easily by a same anesthesiologist, the operation time and the success rate of SGB were similar between groups (Table 1).

Patient comfort

Compared to group A, the patient comfort was better in group B (*P* < 0.001), evidenced by a marked increase in the proportion of comfortable and a concurrent decrease in uncomfortable rates (*P* < 0.05) (Table 2).

Table 2
Comparison of patient comfort and psychological stress between groups

Item	Group A (<i>n</i> = 86)	Group B (<i>n</i> = 95)	χ^2	<i>P</i>
Patient comfort			6.604	0.010
Comfortable	46 (53.5%)	65 (68.4%) *	4.254	0.039
Neutral	31 (36.0%)	28 (29.5%)	0.887	0.346
Uncomfortable	9 (10.5%)	2 (2.1%) *	5.879	0.015
Patient psychological stress			8.722	0.003
Extremely tense	11 (12.8%)	6 (6.3%)	2.241	0.134
Somewhat tense	50 (58.1%)	41 (43.2%) *	4.068	0.044
No tense	25 (29.1%)	48 (50.5%) **	8.744	0.003
Compared with group A, * <i>P</i> < 0.05, ** <i>P</i> < 0.01				

Patient psychological stress

Compared with group A, the patient sin group B exhibited significantly lower emotional stress ($P < 0.05$), characterized by a reduced proportion of somewhat tense and a notable increase in those indicating no tense ($P < 0.01$) (Table 2).

Sleep metrics

With the progression of treatment, both groups demonstrated significant reductions in Pittsburgh Sleep Quality Index (PSQI, group A, $F = 313.677$, $P < 0.001$ vs. group B, $F = 510.775$, $P < 0.001$), Athens Insomnia Scale (AIS, group A, $F = 275.672$, $P < 0.001$ vs. group B, $F = 593.961$, $P < 0.001$), and Hamilton Anxiety Scale (HAMA, group A, $F = 364.843$, $P < 0.001$ vs. group B, $F = 314.975$, $P < 0.001$). No statistically significant intergroup differences were observed for these indicators ($P > 0.05$) (Table 3).

Table 3
Comparison of sleep metrics between groups

Item	Time point	Group A (<i>n</i> = 86)	Group B (<i>n</i> = 95)	<i>F</i>	<i>P</i>
PSQI	T ₀	17.8 ± 2.4	17.6 ± 2.3	0.010	0.524
	T ₁	13.2 ± 3.2	13.7 ± 2.6	5.307	0.240
	T ₂	9.3 ± 3.2	9.6 ± 2.8	2.513	0.492
	T ₃	10.0 ± 3.3	10.2 ± 3.0	0.945	0.739
AIS	T ₀	15.9 ± 3.2	16.3 ± 2.9	0.974	0.441
	T ₁	11.7 ± 3.3	11.4 ± 2.4	5.145	0.490
	T ₂	7.5 ± 2.5	7.6 ± 2.6	0.088	0.726
	T ₃	7.3 ± 2.8	7.2 ± 2.3	2.492	0.922
HAMA	T ₀	16.2 ± 4.5	16.6 ± 4.7	0.295	0.550
	T ₁	12.2 ± 3.6	12.1 ± 3.8	0.001	0.917
	T ₂	9.2 ± 2.9	9.2 ± 2.9	0.609	0.843
	T ₃	9.1 ± 2.6	9.3 ± 2.8	0.011	0.674
AIS, Athens Insomnia Scale; HAMA, Hamilton Anxiety Scale; PSQI, Pittsburgh Sleep Quality Index.					

Insomnia therapy efficacy

The therapeutic efficacy exceeded 80% in both groups, there was no statistically significant difference between groups (*P* > 0.05) (Table 4).

Table 4
Comparison of insomnia therapy efficacy between groups

Item	Group A (<i>n</i> = 86)	Group B (<i>n</i> = 95)	χ^2	<i>P</i>
Insomnia treatment response			0.019	0.990
Markedly effective	32 (37.2%)	36 (37.9%)	0.009	0.924
Effective	38 (44.2%)	41 (43.2%)	0.019	0.889
Invalid	16 (18.6%)	18 (18.9%)	0.003	0.953

Adverse events

No case experienced hypoxemia, nausea, or vomiting. The total number of patients with complications was 62.8% in group A vs. 54.8% in group B ($F = 1.209$, $P = 0.271$). No statistically difference in the incidence of hoarseness, brachial plexus block, dyspnea, dizziness, puncture site pain/bleeding, hypotension, hypertension, bradycardia, and tachycardia was observed between groups ($P > 0.05$) (Table 5).

Table 5
Comparison of adverse events between groups

Item	Group A ($n = 602$)	Group B ($n = 665$)	χ^2	P
Hoarseness	37 (6.1%)	42 (6.3%)	0.016	0.901
Brachial plexus block	26 (4.3%)	27 (4.1%)	0.053	0.818
Dyspnea	18 (3.0%)	18 (2.7%)	0.092	0.762
Dizziness	19 (3.2%)	17 (2.6%)	0.411	0.521
Puncture site pain	5 (0.8%)	7 (1.1%)	0.167	0.683
Puncture site bleeding	7 (1.2%)	8 (1.2%)	0.004	0.947
Hypotension	11 (1.8%)	10 (1.5%)	0.203	0.653
Hypertension	6 (1.0%)	5 (0.8%)	0.220	0.639
Bradycardia	4 (0.7%)	3 (0.5%)	0.262	0.609
Tachycardia	5 (0.8%)	6 (0.9%)	0.019	0.891

Discussion

In recent years, SGB demonstrates significant sleep-enhancing effects through selective preganglionic fiber blockade in the cervicocephalic region, which induces vasodilation of cephalic and upper limb vessels while enhancing regional perfusion¹³. This intervention stabilizes neuronal function by modulating interactions between the cerebral cortex, brainstem ascending reticular activating system, and hypothalamocortical pathways^{4,14}. By preventing excessive polarization of hypothalamocortical projection neurons and reducing norepinephrine release, SGB creates an optimal microenvironment for neurotransmitter secretion in cortical and brainstem nuclei¹⁵. Through inhibition of sleep-regulatory centre excitability, SGB significantly prolongs total sleep duration and improves sleep architecture^{16~18}.

The stellate ganglion, locates at C₇-T₁ level anterior to the vertebral column, is accessible through anterior, side or back approaches regardless of patient positioning^{19,20}. This study utilized ultrasound real-time imaging to guide SGB. Given the superficial location of stellate ganglion, the real-time imaging enabled precise needle placement between the longus colli muscle and carotid artery, minimizing anatomical challenges²¹. Herein all SGB were completed easily, the high success rate (> 98%) across

both groups underscored the reliability of ultrasound guidance for SGB, even in non-standard positions. In addition, the operation time was similar, and there was no significant difference in the incidence of adverse reactions between groups, further indicating that ultrasound-guided seated SGB was safe and feasible.

During invasive procedures, the patient's subjective experience is of critical importance. To enhance patient compliance, this study implemented a gradually tapering treatment regimen. All procedures were conducted on an outpatient basis to minimize two potential confounding factors: tissue trauma from daily needle insertions during hospitalization, and environmental disruption of sleep patterns. While this approach may introduce some degree of treatment discontinuity, it represents a clinically optimized balance between therapeutic efficacy and patient tolerability. The current randomized controlled trial demonstrated that ultrasound-guided sitting SGB provided statistically superior comfort levels and significantly reduced mental stress compared to conventional supine position, as evidenced by validated patient-reported outcomes. These findings corroborate existing evidence suggesting that upright positioning during invasive procedures may enhance procedural autonomy and mitigate anxiety through physiological mechanisms, including optimized proprioceptive feedback and activation of endogenous self-regulatory pathways^{22, 23}; that was particularly beneficial for patients with limited mobility and those with high anxiety or requiring prolonged procedural support (for instance, receiving multi-session injection therapy).

PSQI, AIS, and HAMA are widely utilized in clinical practice to assess patients' sleep quality. PSQI is a self-reported questionnaire evaluating sleep quality over the past month across 7 domains¹⁴, AIS focuses on insomnia severity with 8 items²⁵, HAMA primarily measures anxiety but includes sleep-related items²⁶. Generally, PSQI and AIS are recommended for sleep-specific assessments, while HAMA is supplementary for comorbid anxiety-related sleep disturbances. In this study, post-treatment scores of PSQI, AIS, and HAMA demonstrated significant reduction, and both groups achieved impressive outcome (effective rate exceeded 80%), the comparable efficacy in sleep improvement suggested that ultrasound-guided SGB in the sitting position could be equally effective as the conventional approach.

This research has inherent limitations. Firstly, to enhance the internal validity, we excluded patients with comorbid psychiatric disorders and severe somatic conditions, this may restrict the applicability of our conclusions to real-world patients with complex comorbidities and insomnia. Secondly, the technical reproducibility requires standardized operator training for the sitting position technique. Thirdly, the relatively small sample size and single-centre design may limit the generalizability of our findings. A multi-centre studies with larger sample sizes and more diverse cohorts (including those with psychiatric comorbidities) are warranted to validate and expand upon these findings.

Conclusions

Ultrasound-guided sitting SGB represents a safe and effective alternative to supine positioning for insomnia management, offering superior patient-centred outcomes without compromising clinical

efficacy. This approach merits consideration for clinical implementation, especially in challenging patient populations.

Methods

Study design

This study was registered in the Chinese Clinical Trial Register (<http://www.chictr.org.cn>), retrospectively registered (PID290556), on October 9, 2025. The official registration number will be obtained on November 15, 2025.

Ethics declarations

All procedures performed in studies involving human participants were in accordance with the ethical standards of the Ethical Committee of Shenzhen Third People's Hospital (Shenzhen, Guangdong, China, Approval No. 2024-049 on May 31, 2024) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all eligible participants.

Patient recruitment

From May 2024 to April 2025, patients received SGB for insomnia were recruited. The inclusion criteria were those diagnosed with refractory insomnia; age ≥ 18 years; symptom duration ≥ 6 months; persistent insomnia despite treatment with ≥ 2 distinct modalities (pharmacological, psychotherapeutic, or physical interventions); PSQI > 5 points²⁷. The exclusion criteria were as followed: (1) treatment-related: refusal to complete the full SGB treatment protocol, prior SGB treatment within the preceding 6 months; (2) physiological factors: body mass index > 32 kg/m², uncontrolled systemic conditions (hypertension, diabetes mellitus, hyperthyroidism), significant organ system dysfunction (cardiac, respiratory, hepatic, or renal); (3) reproductive status: pregnancy or planned pregnancy within 6 months; (4) psychiatric considerations: cognitive impairment, history of psychiatric disorders or current psychotropic medication use; (5) safety concerns: known hypersensitivity to local anesthetics. Furthermore, cases involving non-compliant patients who failed to adhere to the prescribed treatment regimen or voluntarily withdrew from the study were also excluded.

Sample size calculation

This was a non-inferiority analysis based on the assumption that the technical difficulty of SGB were comparable between two groups. The sample size was estimated using PASS 23.0 software, and the combined standard deviation of two groups was determined based on the pre-trial results $\sigma = 0.5$, non-inferiority threshold $\delta = 0.2$. With $\alpha = 0.025$ (one side), inspection efficiency $(1 - \beta) = 0.9$, the minimum

sample was calculated as 68 cases for each group. Accounting for an anticipated 20% dropout rate during the study period, a minimum of 82 patients per group was required for enrolment.

Randomization

Patients were divided into two groups based on computer-generated random numbers. If the number was odd, the patients would be assigned to group A (supine position), if the number was even, assigned to group B (sitting position).

Blinding design

It was a prospective, non-blind trial, all SGB was conducted by a single anesthesiologist who was aware of patient grouping, but data was collected and analysed by another anesthesiologist who was not aware of the grouping.

Insomnia therapy protocol

In addition to the patient's ongoing treatment and SGB, all participants received sleep hygiene education and underwent cognitive restructuring for maladaptive sleep beliefs at each visit. Patients were encouraged to maintain a sleep diary and adhere to a structured Cognitive Behavioral Therapy for Insomnia²⁸ program throughout the treatment period.

SGB procedure

After patients entering the treating room, electrocardiograms, non-invasive blood pressure, and pulse oxygen saturation were routinely monitored. In group A, a supine position was taken; the process of ultrasound-guided SGB was as follows: patient's head tilted to one side to expose the contralateral neck, a high-frequency linear array ultrasound probe was used to identify the elliptical hypoechogenic area at C₇ level, between the longus colli muscle and carotid artery. Following the disinfection and draping procedure, an in-plane needle insertion guidance technique was used to target an injection of 0.6% lidocaine 4 ml (Fig. 2). The treatment regimen was as follows: SGB was performed on each side alternately, administered every 1–2 days in the first week (total of 3 times), every 2–3 days in the second week (total of 2 times), once each in the third and fourth weeks, with a total of 7 sessions over 4 weeks. In group B, the same technical parameter was applied except the sitting position (Fig. 3).

Assessment of insomnia therapy efficacy

Patients were stratified into four categories based on the PSQI scores^{27,29,30}: Grade 1 (0–5 points), excellent sleep quality, within normal physiological range; Grade 2 (6–10 points), mild functional disruption, subclinical sleep impairment; Grade 3 (11–15 points), moderate sleep disturbance, clinically significant sleep disorder; Grade 4 (16–21 points), severe sleep impairment, marked daily dysfunction. The therapeutic response to insomnia treatment was categorized as follows: markedly effective, a reduction of ≥ 2 PSQI levels; effective, a reduction of 1 PSQI level; invalid, no change or an increase in PSQI level.

Data collection

The technical difficulty (categorized into easy, moderate, and difficult), the operation time (commencing with ultrasound scanning and concluding with the injection) and the success rate of SGB (Horner syndrome appeared within 30 minutes after injection, presented with miosis, ptosis, enophthalmos, and anhidrosis or hypohidrosis on the affected side³¹), the patient comfort (grade as comfortable, neutral, and uncomfortable), and psychological stress (assessed by the patient, classified as extremely tense, somewhat tense, no tense) during therapy were evaluated. The sleep quality metrics, including PSQI, AIS³², and HAMA³³ were systematically recorded at multiple time points: baseline prior to therapy (T_0), 1 week post-SGB initiation (T_1), 1 month post-SGB initiation (T_2), and 3 months post-treatment (T_3). The effectiveness of insomnia treatment was evaluated. The adverse events such as hoarseness, brachial plexus block, dyspnea, dizziness, puncture site pain or bleeding, hypoxemia, hypotension, hypertension, bradycardia, tachycardia, nausea, and vomiting were also observed.

Outcomes

The primary outcomes were the procedural difficulty and the success rate of SGB. The secondary outcomes were the patient comfort, psychological stress, sleep metrics, and adverse events.

Statistical analyses

Statistical analyses were conducted using SPSS 13.0 (IBM Corp., Armonk, NY, USA). Continuous data were expressed as mean $\pm$ standard deviation ($\bar{x} \pm s$) for normally distributed variables or median (interquartile range [IQR]) for non-normally distributed variables, while categorical data were reported as frequencies and percentages (n [%]). Between-group comparisons for continuous variables were performed using independent samples t-tests for parametric data or *Mann-Whitney U tests* for non-parametric data. Repeated-measures *ANOVA* was employed to analyze time-dependent variations in PSQI, AIS, and HAMA across measurement intervals. Categorical variables were compared using *Pearson's chi-square tests* or *Fisher's exact tests* for small cell counts. Statistical significance was defined as $P < 0.05$ (two-tailed).

Declarations

Acknowledgements

We would like to thank Editage for professional language editing in this study.

Author contributions

Minqiang Liu and Qiang Wu designed the study. Minqiang Liu performed SGB and drafted the manuscript. Liping Qiu collected the data, Aihua Chen and Xiaoliang Cha analysed the data. Qiang Wu revised, and accepted direct responsibility for the manuscript. All authors have approved the final submission.

Data Availability

The datasets used and/or analysed during this study are available from the corresponding author on reasonable request.

Funding

This study was supported by Shenzhen High-level Hospital Construction Fund, Hospital Clinical Projects of Shenzhen Third People's Hospital (25270G1022).

Competing interests

The authors declare no competing interests.

Conflicts of interest

The authors declare no conflicts of interest.

Funding

This study was supported by Shenzhen High-level Hospital Construction Fund and Clinical Projects of Shenzhen Third People's Hospital (25270G1022).

Ethics approval

Approved by the Ethics Committee of Shenzhen Third People's Hospital (No. 2024-049).

Trial registration

Chinese Clinical Trial Register: retrospectively registered (PID290556), on October 9, 2025. The official registration number will be obtained on November 15, 2025.

References

1. Tian, Y. et al. Stellate Ganglion Block Therapy for Complex Regional Pain Syndrome: A Systematic Review and Meta-Analysis. *Pain. Physician.* **27**, 175-184 (2024).
2. Kim, J. et al. Multicenter Prospective Randomized Comparison of Ultrasound-Guided Stellate Ganglion Versus Thoracic Paravertebral Block for Sympathetic Blockade in Chronic Upper Extremity Pain. *Anesth. Analg.* **140**, 665-674 (2025).
3. Liu, Y. et al. Efficacy and safety of stellate ganglion block with different volumes of ropivacaine to improve sleep quality in patients with insomnia: a comparative study. *Eur. Rev. Med. Pharmacol. Sci.* **27**, 10233-10239 (2023).
4. Liu, N. et al. Efficacy and exploratory analysis of potential mechanisms of stellate ganglion block in alleviating sleep disturbance in generalized anxiety disorder: a randomized controlled trial excluding comorbid depression. *Front. Neurol.* **16**, 1554841 (2025).
5. Li, Z. et al. Ultrasound-Guided Stellate Ganglion Block with Cervicotemporal Myofascial Trigger Point Acupuncture for Insomnia with Comorbid Anxiety and Depression: A Retrospective Observational Study. *Nat. Sci. Sleep.* **17**, 2131-2143 (2025).
6. Zhang, X., Huang, X. & Yang, J. Efficacy of stellate ganglion block on postoperative sleep disorder: A systematic review. *Asian. J. Surg.* S1015-9584(24)02117-1 (2024).
7. Luo, J., Zhang, X., Tan, X., Chen, J. & Tang, M. Targeting postoperative sleep disorders: Current advances and emerging directions in stellate ganglion block therapy. *Medicine (Baltimore)*. **104**, e43256 (2025).
8. Park, D.Y. et al. Impact of Neck Position on the Probability of Common Carotid Artery Puncture During Ultrasound-Guided Stellate Ganglion Block. *PM. R.* **11**, 463-469 (2019).
9. Joeng, E.S. et al. Sonoanatomical Change of Phrenic Nerve According to Posture During Ultrasound-Guided Stellate Ganglion Block. *Ann. Rehabil. Med.* **40**, 244-251 (2016).
10. Jeevendiran, A. et al. Comparative Evaluation of Analgesic Efficacy of Ultrasound-Guided Pericapsular Nerve Group Block and Femoral Nerve Block During Positioning of Patients With Hip Fractures for Spinal Anesthesia: A Prospective, Double-Blind, Randomized Controlled Study. *Cureus.* **16**, e56270 (2024).
11. Saxena, P., Singh, M. K., Chaurasia, M. K. & Singh, S. Efficacy of Infraclavicular Brachial Plexus Block Alone Versus Combination With Suprascapular Nerve Block in Patients Undergoing Shoulder Surgeries: A Single-Blind, Randomized Trial. *Cureus.* **16**, e52961 (2024).
12. Mohanty, C. R., Gupta, A., Radhakrishnan, R. V., Singh, N. & Patra, S. K. Ultrasound-guided low-volume anterior suprascapular nerve block for reduction of anterior shoulder dislocation in the emergency department: A case series. *Turk. J. Emerg. Med.* **23**, 254-257 (2023).
13. Feigin, G. et al. Stellate ganglion block for non-pain indications: a scoping review. *Pain. Med.* **24**, 775-781 (2023).
14. Zhou, C. et al. Stellate ganglion block suppresses hippocampal ferroptosis to attenuate cerebral ischemia-reperfusion injury via the Hippo pathway. *Metab. Brain. Dis.* **38**, 1633-1642 (2023).

15. Wang, W. et al. Stellate ganglion block attenuates chronic stress induced depression in rats. *PLoS. One.* **12**, e0183995 (2017).
16. Gu, C. et al. Ultrasound-guided stellate ganglion block improves sleep quality in elderly patients early after thoracoscopic surgery for lung cancer: a randomized controlled study. *J. South. Med. Univ.* **42**, 1807-1814 (2022).
17. Yang, R.Z. et al. Stellate ganglion block improves postoperative sleep quality and analgesia in patients with breast Cancer: a randomized controlled trial. *Pain. Ther.* **12**, 491-503 (2023).
18. Yan, S. et al. Stellate ganglion block alleviates postoperative sleep disturbance in patients undergoing radical surgery for gastrointestinal malignancies. *J. Clin. Sleep. Med.* **19**, 1633-42 (2023).
19. Wu, W., Dai, H., Liu, M., Liu, Y. & Shi, H. Effect of stellate ganglion block on perioperative myocardial injury following thoracoscopic surgery for lung cancer (SGBMI): protocol for a single-centre, randomised controlled trial. *BMJ. Open. Respir. Res.* **11**, e002446 (2024).
20. Motazedian, P. et al. Efficacy of stellate ganglion block in treatment of electrical storm: a systematic review and meta-analysis. *Sci. Rep.* **14**, 24719 (2024).
21. Hou, J., Pu, S., Xu, X., Lu, Z. & Wu, J. Real-time ultrasound-guided stellate ganglion block for migraine: an observational study. *BMC. Anesthesiol.* **22**, 78 (2022).
22. Marmagkiolis, K., Cilingiroglu, M. & Grines, C.L. Transradial Coronary Angiogram in a Sitting Position. *J. Invasive. Cardiol.* **29**, E88-E89 (2017).
23. Amini, M. et al. Percutaneous biopsy of kidney; comparison between prone position and sitting position. *Iran. J. Kidney. Dis.* **13**, 300-303 (2019).
24. Jerković, A. et al. Psychometric Properties of the Pittsburgh Sleep Quality Index (PSQI) in Patients with Multiple Sclerosis: Factor Structure, Reliability, Correlates, and Discrimination. *J. Clin. Med.* **11**, 2037 (2022).
25. Elbi, H. et al. Validation and Discriminant Analysis of the Athens Insomnia Scale in Older Adults. *Int. J. Methods. Psychiatr. Res.* **34**, e70017 (2025).
26. Feng, X. et al. The impact of intradialytic elastic band exercise on physical and cognitive abilities in patients on maintenance hemodialysis: a randomized controlled trial. *Ren. Fail.* **47**, 2482124 (2025).
27. Hinz, A. et al. The Pittsburgh Sleep Quality Index (PSQI) Applied to Cancer Patients: Psychometric Properties and Factors Affecting Sleep Quality. *Cancer. Invest.* **43**, 103-113 (2025).
28. O'Hora, K.P. et al. Comparative effectiveness and predictors of cognitive behavioral therapy for insomnia and its components in older adults: main outcomes of a randomized dismantling trial. *J. Clin. Sleep. Med.* **21**, 1679-1695 (2025).
29. Tan, C. et al. The Limitations of Using the Pittsburgh Sleep Quality Index to Assess Athletes' Sleep Quality: Evidence from Reliability and Validity in Chinese Professional Athletes. *Psychol. Res. Behav. Manag.* **17**, 2603-2617 (2024).

30. Park, S.H. et al. Evaluation of Sleep Disturbance in Alopecia Areata through Questionnaire: Pittsburgh Sleep Quality Index as a Reasonable Tool. *Ann. Dermatol.* **35**, 183-189 (2023).
31. Kim, M.K. et al. Effect of Stellate Ganglion Block on the Regional Hemodynamics of the Upper Extremity: A Randomized Controlled Trial. *Anesth, Analg.* **126**, 1705-1711 (2018).
32. Tang, X. et al. Validation of the Athens Insomnia Scale Among Young Chinese Male Population in a High-Altitude Situation. *Nat. Sci. Sleep.* **16**, 1377-1386 (2024).
33. Liu, Y., Zhu, Z., Yan, J. & Wang, Y. The relationship between anxiety and depression symptoms in insomnia patients: a network analysis. *Sci. Rep.* **15**, 25030 (2025).

Figures

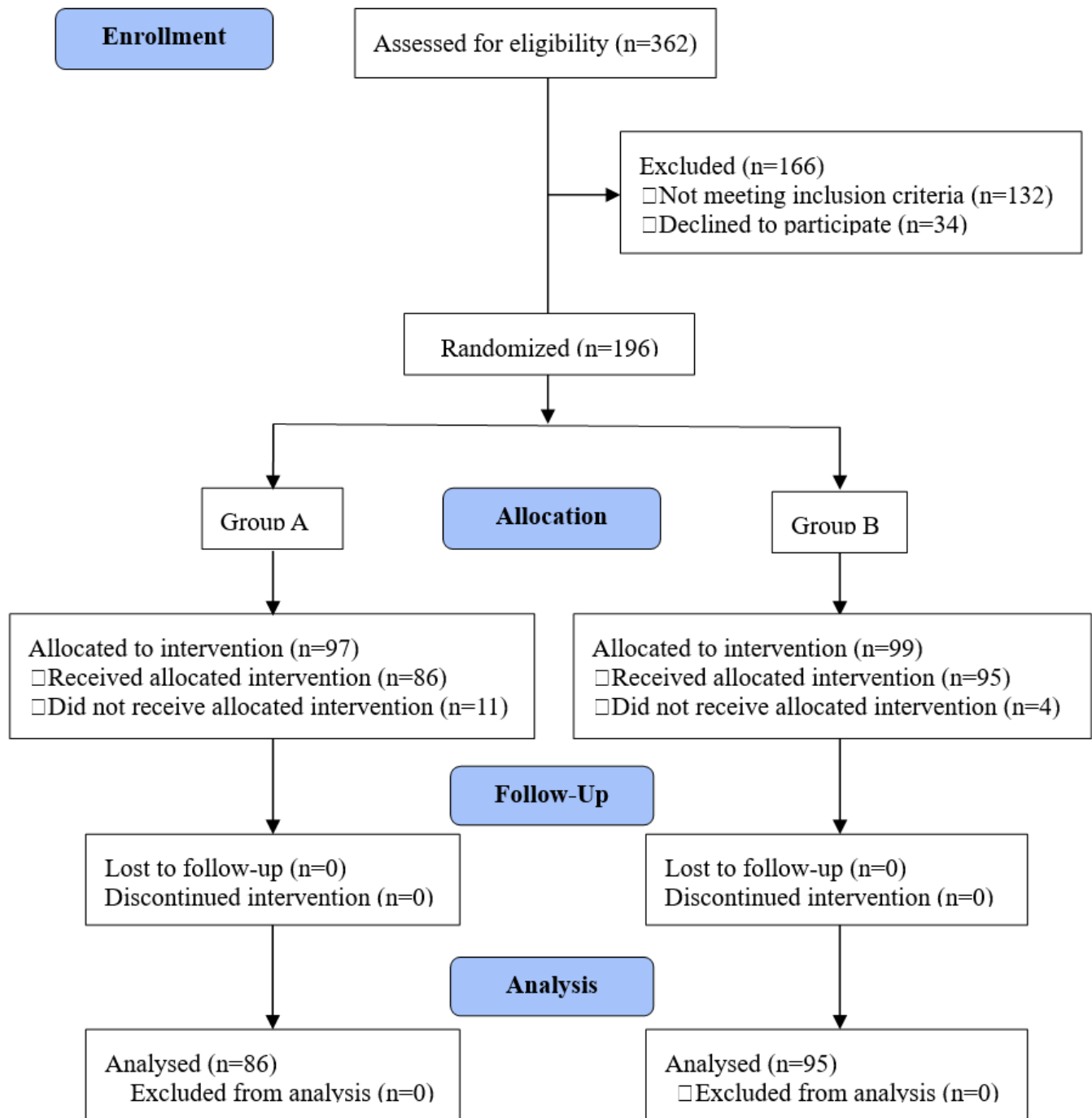


Figure 1

Flow diagram of this study



Figure 2

Ultrasound-guided SGB in supine position



Figure 3

Ultrasound-guided SGB in sitting position.